Initial-state matching and stability diagnostics for a counter-streaming electron-fluid closure

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Abstract

We examine the initial-state matching and closure interpretation in arXiv:2607.07106v1 using its published input decks, archived initial moments, and explicit continuum tests. The central two-population pressure and third-moment identities are correct. However, the PIC input specifies a proper-momentum drift where the fluid input specifies ordinary velocity. In the two archived cells straddling the lower current-sheet center, the density-weighted initial PIC moment current differs by about 11.5% from the magnetostatic current implied by the staggered magnetic curl. Its Cartesian thermal pressures are about 20% below the fluid initialization in p_{xx} and 38.4% below in p_{zz}. A matched-state rerun is needed to determine the effect on the nonlinear comparison. Separately, the stated finite-isotropization closure has an overstable longitudinal mode with growth about $0.00952\omega_{pe}$ in a homogeneous two-Maxwellian equilibrium certified as stable to collisionless longitudinal electrostatic perturbations by the Penrose criterion. We give the linear system, constraint checks, a correction to the symmetry-plane density argument, and a relative heat-flux error bound that accounts for imbalance and cancellation. The calculations distinguish valid mixture algebra, empirical benchmark evidence, and claims requiring additional validation. No new PIC or Gkeyll reconnection simulation is claimed.

1 Introduction and scope

Counter-streaming electron populations provide a useful representation of the strongly non-Maxwellian distributions generated during collisionless magnetic reconnection. McGrae-Menge and collaborators [1] report that tagging electrons by their initial side of a current sheet isolates sub-populations whose intrinsic third moments are small compared with their combined heat flux. Their exact moment decomposition explains how relative motion and population asymmetry can generate a substantial aggregate third moment even when each population has little intrinsic heat flux. Evolving additional population moments is therefore physically motivated.

The present study examines two separate questions that this useful mechanism does not settle: whether the numerical comparison starts from matched physical states, and whether the implemented finite-relaxation closure has the kinetic stability properties attributed to it. These are testable questions about the identified source version and model, rather than an assertion that the underlying mixture interpretation is wrong.

Our first result uses the authors' publicly archived initial moments and input decks [2]. Identical nominal drift expressions enter a proper-momentum initializer in PIC and an ordinary-velocity initializer in the fluid model. The recorded initial moments show a current and pressure mismatch

consistent with that distinction. This is a reproducible control issue for interpreting the nonlinear comparison; its eventual dynamical effect cannot be determined from initial-state analysis alone.

Our second result is a controlled longitudinal stability test of the stated continuum isotropization closure. A positive two-Maxwellian equilibrium is stable to longitudinal electrostatic perturbations under the established Penrose criterion [3], but its finite-relaxation fluid counterpart has an overstable mode. We give an explicit linear system and check both its eigenvalue residual and Gauss constraint. The test is homogeneous and nonrelativistic. It does not replace the source's inhomogeneous reconnection calculation or show that the identified mode caused any of its plotted features.

Two further results make the interpretation more precise. Continuity depends on the divergence of the mass flux, so a zero normal velocity at a symmetry plane does not prohibit density accumulation there. A conditioned third-moment error bound also shows why a thermal-to-drift speed ratio alone does not give a universal relative-error guarantee when population contributions cancel. Neither result disputes the empirical parameter trend in the published scan.

1.1 Relation to established work

Heat-flux closure and its effect on collisionless response are established subjects. Hammett and Perkins [4] developed fluid moment models for Landau damping, while Wang and collaborators [5] compared multifluid moment models with PIC reconnection. Goldman and collaborators [6] studied multibeam energy moments and the distinction between component and aggregate centering. The mixture algebra used below belongs to this established moment framework.

Closure-dependent instability spectra are also known. Ng and collaborators [7] demonstrate why a pressure-tensor model must be checked against kinetic response rather than judged solely by its retained variables. Our finite-relaxation example supplies a concrete test associated with the closure and reference rate in the present source; we do not claim the general phenomenon of closure-dependent unstable modes as new.

Work on heat-flux closure for nonlinear electrostatic two-stream systems [8, 9] further limits a literal reading of broad claims that no fluid closure can capture non-Maxwellian counter-streaming. It does not establish prior use of the specific tagged-population closure in electron-only reconnection. Our analysis therefore addresses explicit initialization, continuity and stability statements, and preserves the source's application-specific empirical evidence.

1.2 Audited version and available evidence

The supplied ten-page PDF is byte-identical to a fresh versioned download of arXiv:2607.07106v1: 7,847,791 bytes, SHA-256 `fc5344e2ba8f4db98baf90d805bb6b7ce 8835c32ea7bf54bba4d62f405b955a3`. The official history checked on 28 October 2026 lists the 8 July v1 and no later revision or journal reference. The PDF displays 9 July, while the experimental HTML displays a different compilation date. The original TeX uses `\date{\today}`; those compilation dates do not identify additional scientific revisions. Appendix A records source and data locators. No original author has been contacted, and no author confirmation or private simulation-build access is claimed.

2 Initial-state matching in the supplied input decks

The source compares a relativistic PIC calculation with nonrelativistic ten-moment fluid calculations and describes their common initialization in Appendices A and C [1]. The deposited reference-

case decks and initial moments make the physical matching of those states directly testable [2]. Equal numerical drift and temperature parameters need not specify equal laboratory-frame velocity moments in the two codes. This issue concerns the controlled comparison; the mixture identities do not depend on either code's initialization convention.

2.1 A proper-momentum parameter is used as a fluid velocity

The PIC deck sets `use_spatial_ufl=.true.` and supplies the curl-derived drift expressions as `spatial_ufl`. The official OSIRIS reference defines these inputs as proper velocities, or momentum divided by particle mass, with relativistic boosting as the default [10]. No classical-addition override appears in the posted deck. The Lua decks instead set nonrelativistic conserved momentum density to $\rho\mathbf{v}$ using the same expressions. This is the ordinary velocity variable of the ten-moment equations [5].

Writing $\mathbf{U} = \mathbf{p}/(m_e c)$, a cold population with proper drift $\mathbf{U}$ has

$$\frac{\mathbf{v}}{c} = \frac{\mathbf{U}}{\sqrt{1+|\mathbf{U}|^2}}. \tag{1}$$

Thus the same numerical vector used for $\mathbf{U}$ in PIC and $\mathbf{v}/c$ in the fluid initializer does not match the physical cold drift. At the unperturbed lower sheet center, the reference expressions give approximately $(0, 0, 0.5)$: the cold PIC speed is $0.4472136c$, whereas the prescribed fluid speed is $0.5c$. The fractional difference is 10.56%. It persists when the thermal spread tends to zero at fixed $B_{\text{Reconn.}} = 0.5$ and sheet width d_e; low thermal beta alone does not remove it. Equation (1) is a cold-limit explanation. A finite-temperature mean must be measured after loading rather than identified with that limit.

2.2 Archived initial current and staggered magnetic curl

The original data authors explicitly define the archived diagnostic `vfl_i` as $(1/n)\int f v_i \,\mathrm{d}^3v$ [2]. It is therefore an ordinary mean velocity, even though the input parameter `ufl` is a proper-momentum parameter. With the source normalization $q_e = -1$, the physical moment current is

$$J_z^{\text{mom}} = -\left(n_1 \overline{v}_{1z} + n_2 \overline{v}_{2z}\right). \tag{2}$$

The selected initial density diagnostics give $n_1 + n_2 = 1$ to within 1.5×10^{-8}. No particle tracking or new time integration is needed to evaluate this current.

The HDF5 files also record their grid offsets: density and velocity moments have $(0.5, 0.5)$, B_x has $(0.5, 1)$ and B_y has $(1, 0.5)$. The backward staggered differences therefore place

$$J^B_{z,i,j} = \frac{B_{y,i,j} - B_{y,i-1,j}}{\Delta x} - \frac{B_{x,i,j} - B_{x,i,j-1}}{\Delta y}, \qquad \Delta x = \Delta y = 0.1 d_e, \tag{3}$$

on the moment-cell centers. Periodic differencing makes the first term's x average exactly zero. The raw data use $y \in [-32, 32] d_e$; the paper's coordinate is $Y = y + 32 d_e$, so its lower sheet is at $Y = 16 d_e$.

The supplied fluid initializer prescribes an x-averaged ordinary velocity $0.49875256c$ in these same cells. The archived PIC means are $0.44098722c$ and $0.44087669c$, approximately 11.6% smaller. The small difference between this continuum prescription and Table 1's staggered curl does not explain the observed deficit. The adjacent cells at the other sheet show the same effect with reversed current sign.

Table 1: Initial x-averaged currents in the cells adjoining the lower sheet. The deficit is $1 - \langle J_z^{\mathrm{mom}}\rangle_x / \langle J_z^B\rangle_x$. Currents use the source's normalized units.

Y/d_e	$\langle J_z^{\mathrm{mom}}\rangle_x$	$\langle J_z^B\rangle_x$	Deficit
15.95	−0.44098723	−0.49833998	11.5088%
16.05	−0.44087669	−0.49833998	11.5309%

Here J_z^{mom} is a moment-based physical diagnostic. We have not reloaded OSIRIS's native charge-conserving deposited current, which can differ from a simple qv deposition. These numbers therefore do not specify the exact discrete electric-field update of the original executable. In the normalized continuum equations, $\partial_t E_z = (\nabla \times \mathbf{B})_z - J_z$. The comparison therefore measures initial magnetostatic balance and a possible electric-field transient, while the field remains free to evolve.

2.3 The laboratory-frame pressure moments also differ

The record defines its second-moment diagnostics without a density normalization: $M_{s,ii} = \int f_s v_i^2 \, \mathrm{d}^3 v$ [2]. Following the source's own pressure definition, at $m_e = 1$ we compute

$$P_{ii} = M_{1,ii} + M_{2,ii} - n\overline{v}_i^2. \tag{4}$$

The posted fluid initializers prescribe an isotropic thermal pressure

$$p_{\mathrm{fl}} = n_0 (0.111803)^2 = 0.012499910809.$$

This is the pointwise initializer prescription; the archived fluid `.bp` fields have not been reloaded. The measured ratios are

Y/d_e	$\langle P_{xx}\rangle_x / p_{\mathrm{fl}}$	$\langle P_{zz}\rangle_x / p_{\mathrm{fl}}$
15.95	0.795900	0.615727
16.05	0.798040	0.615447

For example, at $Y = 15.95 d_e$ the archived values are $\langle P_{xx}\rangle_x = 0.00994868$ and $\langle P_{zz}\rangle_x = 0.00769653$. The latter is about 38.4% below the fluid pressure prescription. These are the ordinary-velocity covariance moments defined in the source, not components of a relativistic energy–momentum tensor. A common rest-frame thermal-momentum parameter does not imply equal laboratory-frame velocity covariance after a boost. Figure 1 shows the full x-averaged profiles.

2.4 What a controlled comparison requires

The data establish an initial-state mismatch in the posted comparison. They do not establish how much of any later reconnection difference it causes. That requires reruns with controlled initial moments and numerical refinement. One constructive test is to initialize the fluid models from the measured PIC n, $\overline{\mathbf{v}}$ and P_{ij}, preserving the intended population labels. Another is to calibrate the PIC loading to a prescribed physical fluid state and then verify the deposited moments. The cold conversion in Eq. (1) is necessary for a cold drift match, but by itself does not guarantee a finite-temperature pressure or current match.

An exact executable build or commit identifier was not found in the inspected deposit. The current public OSIRIS code inspected for documentation does not expose the authors' custom `vfl`/third-moment diagnostic implementation. Our diagnostic interpretation follows their explicit

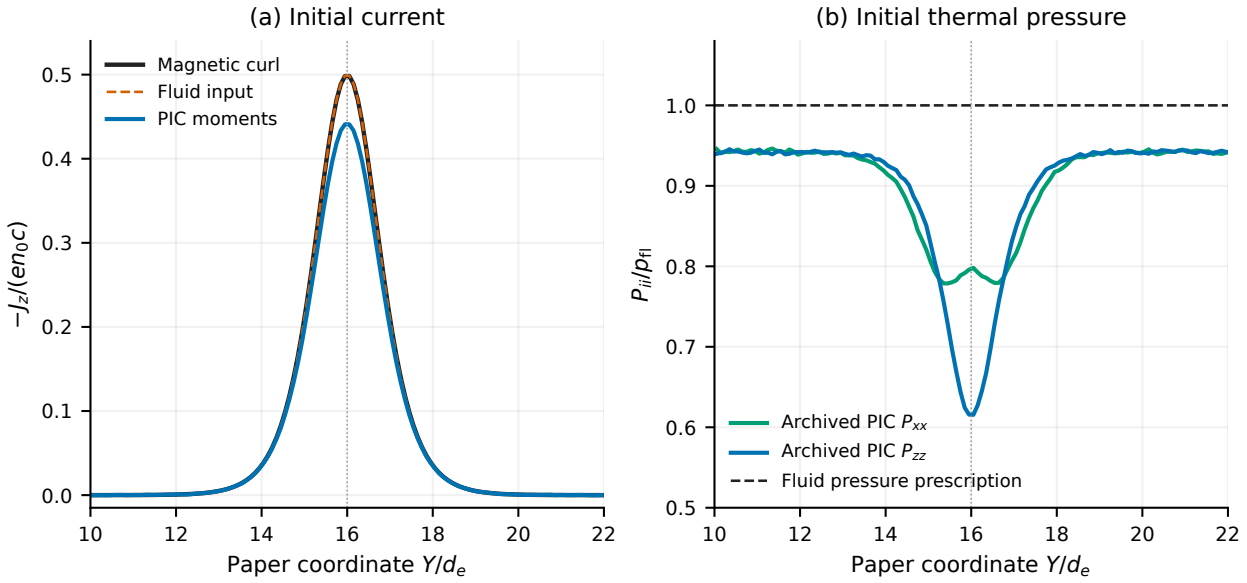


Figure 1: Reanalysis of the archived initial state near the lower sheet. Left: the ordinary-velocity moment current differs from both the staggered magnetic curl and the posted fluid prescription. Right: the local centered pressure components, averaged in x after centering, differ from the isotropic fluid prescription. The profiles use the authors' archived diagnostics and their published definitions [2]. The fluid curves are initializer prescriptions, not reloaded fluid output. No new simulation is shown.

dataset definitions and file metadata. All selected files are at $t = 0$. The present calculation is a reproducible reanalysis of those files, with no new PIC or Gkeyll run and no causal attribution of the source's later nonlinear results.

3 A controlled linear-stability test

The source states that resolving counter-streaming populations allows its two-fluid model to capture the linear growth of the associated instabilities [1]. Population resolution alone does not guarantee the correct kinetic response. Closure-dependent differences in fluid and kinetic instability spectra are established issues [7, 5]. We give an explicit test of the stated local-isotropization closure, without inferring that the test mode occurs in the source's nonlinear reconnection calculation.

Consider a homogeneous nonrelativistic electron plasma with immobile neutralizing ions. Two equal-density populations have ordinary velocities $\pm U e_x$ and isotropic thermal variance σ^2. Their longitudinal velocity marginal, normalized to unit density, is

$$f_0(v) = \frac{e^{-(v-U)^2/(2\sigma^2)} + e^{-(v+U)^2/(2\sigma^2)}}{2\sqrt{2\pi}\sigma}. \tag{5}$$

A constant magnetic field parallel to e_x leaves the longitudinal, gyrotropic perturbation subspace used below invariant.

The collisionless kinetic reference. For $U > \sigma$, the only local minimum of f_0 is at zero. Indeed, $f_0'/f_0 = [-v + U\tanh(Uv/\sigma^2)]/\sigma^2$, and $U\tanh(Uv/\sigma^2) - v$ is strictly concave on $v > 0$,

with exactly one positive zero. The Penrose criterion [3] therefore tests

$$I = \int_{\mathbb{R}} \frac{f_0'(v)}{v}\,\mathrm{d}v = \frac{2\zeta D(\zeta) - 1}{\sigma^2}, \qquad \zeta = \frac{U}{\sqrt{2}\sigma}, \quad D(z) = e^{-z^2}\int_0^z e^{t^2}\,\mathrm{d}t. \tag{6}$$

This integrand has a removable singularity at zero. At $U/\sigma = 6/5$, $\zeta^2 = 18/25$. The alternating Dawson series has decreasing term magnitudes and gives the rational upper bound

$$2\zeta D(\zeta) \le \frac{36}{25} - \frac{432}{625} + \frac{15552}{78125} = \frac{74052}{78125} < 1. \tag{7}$$

Consequently $I < 0$, and this kinetic equilibrium has no exponentially growing longitudinal electrostatic mode for any real wavenumber. Direct quadrature gives $\sigma^2 I = -0.08730405848\ldots$; the sign does not depend on that numerical evaluation.

Linearizing the finite-relaxation fluid model. The source retains $\partial_m q_{sijm} = \nu(p_{sij} - \delta_{ij}\,\mathrm{tr}\,p_s/3)$, with $\nu = \kappa\sigma > 0$. Normalize the total plasma frequency, density, electron mass, charge magnitude and permittivity to one. For each beam $s = \pm$, set $U_s = sU$, $n_s = 1/2$, and equilibrium diagonal pressure $p_s^0 = n_s\sigma^2$. Write its perturbations as $N_s = \delta n_s$, $V_s = \delta u_{sx}$, $P_s = \delta p_{sxx}$ and $H_s = \delta\,\mathrm{tr}\,p_s$. For spatial factor e^{ikx}, the invariant nine-variable system is

$$\begin{aligned}
\dot N_s &= -ikU_sN_s - ikn_sV_s,\\
\dot V_s &= -ikU_sV_s - ikP_s/n_s - E,\\
\dot P_s &= -ikU_sP_s - 3ikp_s^0V_s - \nu(P_s - H_s/3),\\
\dot H_s &= -ikU_sH_s - 5ikp_s^0V_s,\\
\dot E &= \sum_{s=\pm}(U_sN_s + n_sV_s).
\end{aligned} \tag{8}$$

Here $\delta p_{syy} = \delta p_{szz} = (H_s - P_s)/2$. A perturbation of ν multiplies the zero equilibrium deviator and thus contributes nothing at linear order, even when thermal speed is computed locally. Eliminating these variables independently, with temporal convention $e^{-i\omega t}$, gives

$$\begin{aligned}
\epsilon_{\mathrm{fl}}(\omega,k) &= 1 - \frac{1}{2}\sum_{s=\pm}\frac{1}{(\omega - kU_s)^2 - k^2\sigma^2 g_s},\\
g_s &= \frac{5}{3} + \frac{4D_s}{3(D_s+\nu)}, \qquad D_s = -i(\omega - kU_s).
\end{aligned} \tag{9}$$

Proposition 3.1 (A finite-relaxation false positive)**.** *At $U/\sigma = 6/5$ and $k\sigma/\omega_{pe} = 1/5$, the collisionless kinetic equilibrium is stable to longitudinal electrostatic perturbations, while the local-isotropization fluid model is overstable for all sufficiently small positive ν.*

Proof. Kinetic stability follows from (7). At $\nu = 0$ the longitudinal fluid pressure response has index three. Putting $W = \omega^2$, $a = k^2U^2$, $b = 3k^2\sigma^2$, its dispersion equation is

$$W^2 - [2(a+b)+1]W + (a-b)(a-b-1) = 0. \tag{10}$$

In units $\sigma = \omega_{pe} = 1$, its simple lower positive frequency is $\omega_0 = [(847 - 5\sqrt{24553})/1250]^{1/2} = 0.2254420613\ldots$. Let $\Omega_s = \omega_0 - kU_s$ and $A_s = \Omega_s^2 - 3k^2$. Implicit differentiation of (9) yields

$$\left.\frac{\mathrm{d}\,\mathrm{Im}\,\omega}{\mathrm{d}\nu}\right|_{\nu=0} = -\frac{2k^2}{3}\frac{\sum_s 1/(\Omega_s A_s^2)}{\sum_s \Omega_s/A_s^2} = 2.488846777\ldots > 0. \tag{11}$$

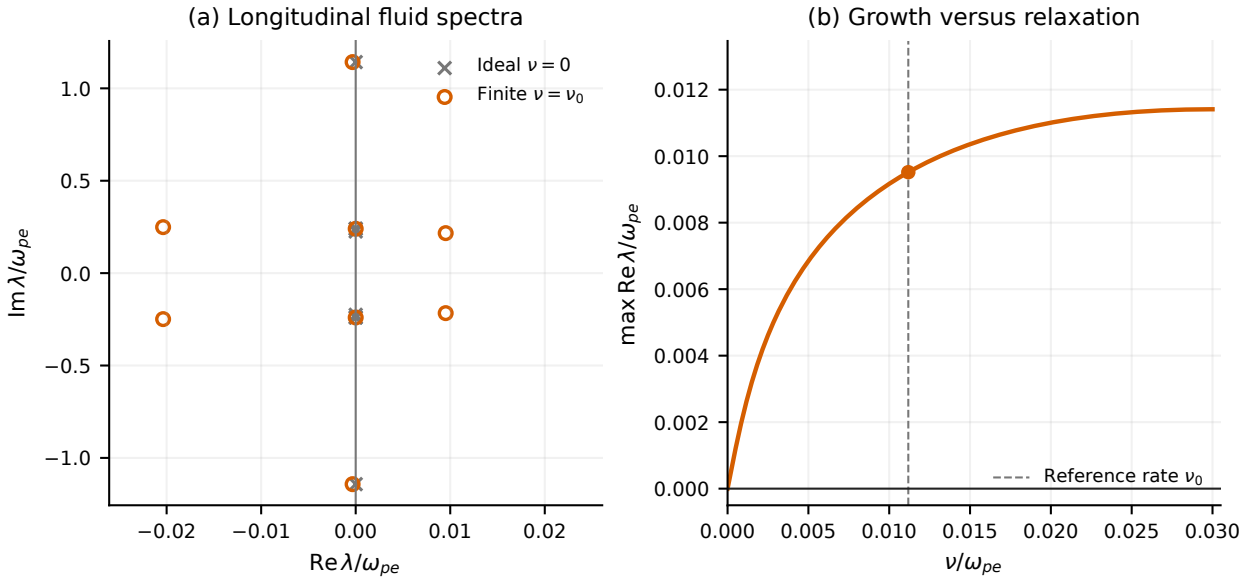


Figure 2: Homogeneous longitudinal test with $U/\sigma = 1.2$ and $k\sigma/\omega_{pe} = 0.2$. Left: nonzero eigenvalues of the ideal and finite-rate fluid systems; the static Gauss-constraint mode is excluded. Right: the largest fluid growth rate as relaxation varies, with $\nu_0/\omega_{pe} = 1/(40\sqrt{5})$ marked. The collisionless kinetic reference has no growing longitudinal electrostatic mode by the Penrose proof above. This comparison concerns the stated continuum closure, not an observed mode in the original reconnection calculation.

The sign can also be checked without the decimal value: the rational bounds $0.225 < \omega_0 < 0.226$ make the numerator sum negative and the denominator sum positive. The root is simple, so its continuation has positive imaginary part for sufficiently small positive relaxation. □

At a reference-scale rate obtained from $\kappa = 0.1/d_e$ and $\sigma = c/\sqrt{80}$, $\nu/\omega_{pe} = 1/(40\sqrt{5}) = 0.01118033989\ldots$, the matrix in (8) numerically gives

$$\frac{\lambda}{\omega_{pe}} = 0.00951849926 \;\pm\; 0.21649095264\,i, \qquad \lambda = -i\omega. \tag{12}$$

The eigen-equation, Gauss constraint $ikE + N_+ + N_- = 0$, and independent dielectric residual are all below 2×10^{-10} in the recorded checks; actual residuals are much smaller. The zero-relaxation matrix has no positive growth beyond roundoff. The literal fluid deck rounds its thermal speed to $0.111803c$; substituting the resulting $\nu/\omega_{pe} = 0.0111803$ also leaves positive growth, with a change smaller than $10^{-6}\omega_{pe}$. A box-compatible check using mode 14 of $L_x = 48d_e$, with $\sigma = c/\sqrt{80}$, gives growth $0.00951299405\omega_{pe}$ and a wavelength resolved by about 34 cells at the reported grid spacing.

These are continuum model diagnostics. They establish a false positive relative to the specified collisionless kinetic reference, including at the source's positive relaxation-rate scale. They do not identify a mode measured in the original current sheet, replace a kinetic stability analysis of that sheet, or show that a particular nonlinear plot is contaminated.

4 Density can grow at a symmetry-plane stagnation point

On PDF p. 5, the source attributes its one-fluid density deficit to the vanishing of u_y at the symmetric mid-plane, describing this as blocking transport into the sheet [1]. The value of a flux

at one plane does not determine its divergence:

$$\partial_t n = -\partial_x(nu_x) - \partial_y(nu_y). \tag{13}$$

In particular, $\partial_y(nu_y)$ can be nonzero where $u_y = 0$.

Proposition 4.1 (A periodic density-accumulation example)**.** *For $a, n_0 > 0$, on a periodic coordinate $y \in [0, 2\pi)$,*

$$u_y(y) = -a \sin y, \qquad n(y,t) = \frac{n_0}{\cosh(at) - \cos y \, \sinh(at)} \tag{14}$$

is a smooth positive solution of the one-dimensional continuity equation. Its total mass is $2\pi n_0$, and $u_y(0) = 0$ while $n(0,t) = n_0 e^{at}$.

Proof. The denominator is positive for every finite t. Differentiating (14) gives $\partial_t n + \partial_y(nu_y) = 0$ directly. The standard integral $\int_0^{2\pi}(A - B\cos y)^{-1}\mathrm{d}y = 2\pi/(A^2 - B^2)^{1/2}$, with $A = \cosh(at)$ and $B = \sinh(at)$, gives the conserved mass. Evaluation at zero gives the stated increasing central density. □

This is an exact kinematic transport example, not a self-consistent plasma–Maxwell solution or a substitute reconnection run. It refutes the inference from symmetry alone. The two-population model's total density and density-weighted velocity obey exactly the same continuity equation after the population equations are summed. A one-fluid model loses the separate population-crossing information, but it is not geometrically forbidden from supplying or accumulating total density within a finite-width layer. Differences in the evolution of aggregate pressure and higher moments can still explain a particular benchmark failure.

5 Conditioning a relative heat-flux accuracy claim

The source's Eqs. (3)–(4) and (B1)–(B2) are correct. For $n_1, n_2 > 0$, put $n = n_1 + n_2$, $w_s = n_s/n$, $\Delta u = u_1 - u_2$ and $T_s = p_s/n_s$. Recentring each population on the aggregate mean gives

$$\begin{aligned} p_{ij} &= p_{1ij} + p_{2ij} + mnw_1w_2\Delta u_i\Delta u_j, \\ q_{ijk} &= q_{1ijk} + q_{2ijk} + nw_1w_2(\Delta u_i\Delta T_{jk} + \Delta u_j\Delta T_{ik} + \Delta u_k\Delta T_{ij}) \\ &\quad - mnw_1w_2(w_1 - w_2)\Delta u_i\Delta u_j\Delta u_k. \end{aligned} \tag{15}$$

The source explicitly defines the unnormalized three-term symmetrization, so its tensor formula has the required scalar factor three. Direct positive finite-mixture calculations, including nonunit mass, verify all entries.

For the scalar xxx component and $\Delta u \neq 0$, define $\theta_s = T_{sxx}/[m(\Delta u)^2]$. Its counter-streaming contribution is

$$q_{\mathrm{CS}} = mn(\Delta u)^3 C, \qquad C = w_1w_2\{3(\theta_1 - \theta_2) - (w_1 - w_2)\}. \tag{16}$$

Thus the source's cubic-drift explanation of the empirical parameter collapse requires qualification. Equation (16) also contains a temperature-contrast term linear in drift and can vanish by symmetry or cancellation.

Proposition 5.1 (A sufficient conditioned relative-error bound)**.** *Suppose $|q_{sxxx}| \le mn_s a_s v_{\mathrm{th},s}^3$, with $v_{\mathrm{th},s} = (T_{sxx}/m)^{1/2}$ and stated nonnegative bounds a_s. Set $E = \sum_s w_s a_s (v_{\mathrm{th},s}/|\Delta u|)^3$. If $E < |C|$, then*

$$\frac{|q_{xxx} - q_{\mathrm{CS}}|}{|q_{xxx}|} \le \frac{E}{|C| - E}. \tag{17}$$

Proof. The exact residual is $r = q_{1xxx} + q_{2xxx}$, so $|r| \le mn|\Delta u|^3 E$. The reverse triangle inequality gives $|q_{xxx}| \ge mn|\Delta u|^3(|C| - E) > 0$. Division gives (17). □

The bound is sufficient rather than necessary. Its imbalance/cancellation factor cannot be dropped. Equal population densities and temperatures give $q_{\mathrm{CS}} = 0$ at any drift. With equal densities but unequal temperatures, $q_{\mathrm{CS}} = 3n\Delta u\,\Delta T_{xx}/4$, which is linear in drift.

A near-Maxwellian counterexample to a speed-ratio-only guarantee. Let W take values $-1/\sqrt{2}$ and $\sqrt{2}$ with probabilities 2/3 and 1/3. Then $\mathbb{E}[W] = 0$, $\mathbb{E}[W^2] = 1$ and $\mathbb{E}[W^3] = 1/\sqrt{2}$. For $\eta > 0$, let $S = (W + \eta G)/\sqrt{1+\eta^2}$ with G a standard Gaussian. Let $H = (S, G_y, G_z)$, with independent standard Gaussian transverse components, and define X_ϵ to have law $(1-\epsilon)\mathcal{N}(0, I) + \epsilon\mathcal{L}(H)$, $0 < \epsilon < 1$. It has zero mean, covariance I, and third moment $\mathbb{E}\Big[X^3_{\epsilon,x}\Big] = \epsilon/[\sqrt{2}(1+\eta^2)^{3/2}]$. Its total-variation distance from $\mathcal{N}(0, I)$ is at most ϵ. Give the populations equal densities and velocity laws

$$V_1 = Ue_x + \sigma X_\epsilon, \qquad V_2 = -Ue_x + \sigma X_\epsilon. \tag{18}$$

Each population density is a positive smooth Gaussian mixture, within ϵ in total variation of its matching Maxwellian. Their intrinsic temperature tensors are both $m\sigma^2 I$, so $q_{\mathrm{CS}} = 0$, whereas $q_{xxx} = mn\sigma^3\epsilon/[\sqrt{2}(1+\eta^2)^{3/2}] \ne 0$. Consequently the relative closure error is one even when both ϵ and the thermal-to-beam speed ratio σ/U are arbitrarily small. The absolute residual does tend to zero relative to mnU^3. The recorded calculations use $\epsilon = 0.01$ and integrate the smooth laws with positive Gauss–Hermite quadrature.

This construction does not disprove the source's measured collapse in its particular family of reconnection simulations. It identifies assumptions needed to extend that empirical observation to a relative-error guarantee. The source scan uses initial speeds; inserting those into a local bound would also require estimates of their subsequent evolution.

The valid limitation of a memoryless one-fluid closure. Additional population information can be genuinely necessary. For example, let $n = m = 1$, and let each population be an isotropic Gaussian of variance $3I/4$. Choose a weight w, set $\Delta = [4w(1-w)]^{-1/2}$, and assign means $(1-w)\Delta e_x$ and $-w\Delta e_x$. For both $w = 1/2$ and $w = 1/4$, the aggregate density is one, the mean is zero, and the pressure is $\mathrm{diag}(1, 3/4, 3/4)$. The respective q_{xxx} values are zero and $1/(4\sqrt{3})$. Hence no arbitrary-distribution algebraic function of only n, u, p can recover both third moments. This supports the source's population-resolving motivation. It is not an impossibility statement for one-fluid descriptions with memory, extra variables, gradients or nonlocality.

6 Finite isotropization over a comparison interval

The source openly retains isotropization for numerical stability [1]. Its smallness relative to a fast local heat-flux scale does not establish closeness to zero-intrinsic-heat-flux evolution over an entire simulation. A homogeneous calculation makes the distinct finite-time parameter explicit.

At fixed density, let p be a positive-definite pressure tensor and $\bar{p} = \operatorname{tr} p/3$. The relaxation part alone satisfies

$$\dot{p} = -\nu(p - \bar{p}I), \qquad p(t) = \bar{p}I + e^{-\nu t}\big(p(0) - \bar{p}I\big). \tag{19}$$

Its trace, and thus its thermal energy, are constant. For the Gaussian entropy diagnostic per particle, $s_G = \frac{1}{2}\log\det p - \frac{5}{2}\log n + \text{constant}$, one obtains

$$\dot{s}_G = \frac{\nu}{2}\big(\bar{p}\,\operatorname{tr}(p^{-1}) - 3\big) \ge 0, \tag{20}$$

Table 2: Selected verification outcomes. The code, data definitions and scope limits distinguish initial-state reanalysis from independent continuum tests.

Check	Outcome
Initial moment current at $Y = 16.05d_e$	11.5309% below staggered curl
Initial P_{zz}/p_{fl} at that row	0.615446655
Exact mixture identities	All scalar and tensor checks pass
Symmetry-plane continuity	Zero normal velocity; increasing density
Conditioned relative-error example	$0.024134 \leq 0.036243$
Identical one-fluid lower moments	Different third moments
Finite-relaxation mode growth	$0.009518499\,\omega_{pe}$
Homogeneous relaxation over $300/\omega_{pe}$	3.4941% anisotropy remains

with equality only at isotropy. The inequality follows by applying the arithmetic–harmonic mean inequality to the three positive eigenvalues of p. This is an entropy diagnostic of the moment model, not a measurement of PIC entropy or evidence of external heat input.

The deviation from the zero-relaxation homogeneous pressure is controlled by $1 - e^{-\nu T}$, not by ν alone. At the ideal reference-scale rate $\nu/\omega_{pe} = 1/(40\sqrt{5})$ and $T = 300\omega_{pe}^{-1}$,

$$\nu T = 3.354101966\ldots, \qquad e^{-\nu T} = 0.0349407340\ldots. \tag{21}$$

Thus an initially present homogeneous anisotropy would retain about 3.49% of its amplitude over that interval. For the example $p(0) = \mathrm{diag}(2, 1/2, 1/2)$ at unit density, the entropy diagnostic increases by $0.3461208970\ldots$ while the trace remains three.

This example does not establish how much relaxation changes the evolving source current sheet, whose anisotropy is time-dependent. It shows why a finite-duration sensitivity study in κ, or a direct comparison of the retained pressure-source terms, is needed when interpreting the implementation as approximately zero-heat-flux evolution. It does not dispute the source's disclosed need for finite stabilization, nor identify irreversible adiabatic relaxation with external energy transfer.

7 Reproducible calculations and evidence limits

The repository separates reanalysis of archived simulation output from analytical model tests. The initial-state analysis uses the authors' data dictionary, independently readable HDF5 attributes, published input decks, and explicitly aligned grid locations. The stability, continuity and moment calculations use specified equations and inputs and require no reconnection simulation executable.

The data retriever requests only selected ZIP members from the public Zenodo archive. It requires bounded HTTP partial responses, checks member lengths and CRC32 against the ZIP directory, and records SHA-256 hashes of the extracted inputs. The published checksum of the complete archive is identified as publisher-supplied; a partial retrieval does not verify that whole-archive checksum. Small derived profiles and provenance are included with their attribution. The original paper and full simulation archives are not redistributed by this project.

Numerical checks preserve the exact mixture identities rather than treating them as errors. The linear calculation is checked independently through an eliminated dielectric relation and the Gauss constraint, in addition to the matrix eigenvalue residual. The kinetic comparison has an exact rational stability bound, so its conclusion does not depend on locating a numerical threshold near marginal stability. The continuity example satisfies the transport equation explicitly. These checks test the scientific statements used in the paper, not only whether a plotting script runs.

The execution record in Appendix B pins the code, input hashes, numerical results and environment. Its timestamps are author-environment execution evidence, not third-party certification or independent human peer review. The accompanying verification script reports actual failures instead of silently replacing absent inputs with synthetic data.

No OSIRIS or Gkeyll reconnection run was performed for this study. The published input decks and archived diagnostics do not uniquely identify the authors' exact executable builds. In particular, the documented custom velocity diagnostics are interpreted using the authors' explicit data dictionary; we do not assert that a present public source checkout is the private build that produced them. This limits claims about implementation internals while leaving the documented archived-moment comparison testable.

8 Consequences for controlled validation

An initialization audit should precede attributing every later difference to the heat-flux closure. The current test exposes a mismatch in physical moments despite matching algebraic drift expressions. A controlled repeat should match density, ordinary mean velocity and the selected pressure definition, and report the initial magnetic curl and moment current on their actual staggered locations. If relativistic particle initialization and a nonrelativistic fluid model are compared, the conversion and the target moment convention must be stated explicitly. The cold proper-velocity conversion alone is not a proof that finite-temperature moments have been matched.

The measured initial mismatch does not determine the sign or size of its effect at the reported $t = 300\omega_{pe}^{-1}$ snapshot. Reconnection may relax some initial differences, amplify others, or preserve the qualitative advantage of retaining two populations. A matched-initial-state comparison and resolution or relaxation-rate studies are needed to separate those possibilities. We therefore do not infer that correcting the drift would eliminate the reported single-fluid density deficit, reproduce all PIC heat flux with one fluid, or reverse the source's qualitative conclusion.

The homogeneous stability test supplies a separate validation requirement. Retaining population drift creates the possibility of streaming modes; it does not ensure the correct kinetic susceptibility or growth rate. The implemented relaxation should be included in the dispersion calculation. Finite relaxation can change a spectrum even when the unperturbed pressure is isotropic and the relaxation source vanishes at equilibrium. In the stated example the resulting growing mode is a false positive relative to the collisionless kinetic reference. This is a test of a model's continuum response, not evidence that the original simulation visibly contains that mode.

The pressure-source diagnostic should also distinguish a third-moment amplitude from its divergence. The scale proxy $p\,d_e/|q|$ used in the source is a useful estimate when d_e is an appropriate gradient length. It is not identical to $p/|\nabla\cdot q|$. A spatially constant nonzero heat-flux tensor has a finite amplitude proxy and zero divergence. The source explicitly notes that its isotropization closure constrains a divergence and omits the reconstructed fluid tensor from some plots; we do not characterize that disclosure as a hidden omission. Direct source-term comparisons and rate convergence would make the approximation more informative.

Finally, the source's time-maximum spatial score is a best-time diagnostic, not a uniform-in-time error bound or a forward prediction test at every scan point. The conditioned estimate above explains why both absolute residuals and a nondegenerate denominator are needed when assessing relative accuracy. These qualifications preserve the observed scan and suggest measurements needed before extending its interpretation to new current sheets, later island recirculation, or turbulence.

The valid component-to-aggregate moment decomposition remains the foundation of this analysis. Additional labels carry information absent from a generic state described only by total density,

velocity and pressure. The corrections identify controls and assumptions needed for using that additional information reliably, rather than a general rejection of multicomponent fluid modeling.

9 Conclusion

Matching a nominal drift expression does not match a proper-momentum PIC initialization to an ordinary-velocity fluid initialization. The archived initial moments make that distinction quantitatively testable. Resolving two populations also does not ensure faithful kinetic instability growth: finite isotropization must be validated as part of the closure. The exact moment decomposition remains valid and useful. Correcting the density-transport explanation and conditioning relative-error statements provides a constructive basis for further comparison, without asserting that the reported nonlinear reconnection results have been disproved.

Author and methods disclosure

Taylor Kalin is the author and is responsible for this study. OpenAI Codex assisted with source inspection, derivations, code, data analysis, and manuscript preparation under his direction. Separate AI-agent checks were used during preparation; they are not represented as independent human expert review. Reported results are deterministic classical computations and reanalysis of public archived moments. No new reconnection simulation or experimental measurement is claimed.

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Prior dissemination and review sequencing

This is one study. GitHub and AiraXiv deposits are preprint versions of the same work. Agent-only editions, where used, identify Taylor Kalin as canonical author and human operator; they are not independent new findings. Review venues are sequenced to avoid concurrent consideration of the same manuscript. Actual identifiers and outcomes are recorded at https://github.com/Teylersf/counterstreaming-heatflux-audit in the publication-status record.

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A Source passages and artifact scope

All source equation and page locators refer to the PDF of arXiv:2607.07106v1 [1]. The supplied ten-page PDF is byte-identical to a fresh versioned arXiv download: 7,847,791 bytes, SHA-256

```
fc5344e2ba8f4db98baf90d805bb6b7c
e8835c32ea7bf54bba4d62f405b955a3.
```

The official history checked on 28 October 2026 lists v1 submitted 8 July 2026 at 07:43:25 UTC, with no later revision or journal reference. The PDF displays 9 July; the experimental HTML displays 24 August. The original `apssamp.tex`, line 95, uses `\date{\today}`, so separately compiled display dates are not version identifiers. No source author has been contacted, and no author confirmation is claimed.

The separately retrieved versioned source archive has 7,439,982 bytes and SHA-256

```
4f9abd5f41ffca0f1879981429165ff1
a989276792c812e860f57cb5038c3b2e.
```

The repository's selective excerpts contain six attributed files and 17 fragments, with original TeX line and character positions. A standalone verifier checks the archive and source hashes and reconstructs every excerpt without executing the source or making network requests. The full PDF and source archive are not redistributed.

The stated initialization. Appendix A, p. 6, states that the electron fluid velocities are initialized from

$$\epsilon_{ijk}\partial_j B_k = J_i = q_e n_0 \overline{v}_i.$$

Appendix C, p. 8, states that the fluid simulations inherit the PIC domain, grid, boundary conditions and magnetic-field initial condition exactly. It specifies the common initial drift

$$\overline{v}_{1i} = \overline{v}_{2i} = \overline{v}_i$$

and thermal speed $0.111803c$ for the electron fluids. These are the matching statements tested in the main text. Original TeX locators are lines 225, 244 and 280; the printed seed is Eq. (A3), lines 238–240.

The current public `udist` documentation describes `ufl` and `spatial_ufl` as fluid momentum in proper-velocity units and gives relativistic momentum addition as the default [10]. This documents the input-language interpretation; it does not identify the authors' exact historical executable. For the archived diagnostics, the definition used here comes directly from the authors' Zenodo deposit description [2]:

$$\mathrm{vfl}_i = \frac{1}{n}\int f v_i \,\mathrm{d}^3 v, \qquad (v_i v_j)_{\mathrm{diagnostic}} = \int f v_i v_j \,\mathrm{d}^3 v.$$

The deck and diagnostic checks, rather than an assumed private code revision, support the initial-moment comparison.

The density-transport explanation. The main text, p. 5, attributes the one-electron-fluid density deficit to the following mechanism:

> "at the mid-plane $\overline{v}_y = 0$ by symmetry, blocking 1-e-Fluid transport into the sheet".

The complete argument is in original TeX line 191. The symmetry-plane continuity calculation assesses this causal explanation; the plotted density deficit remains empirical evidence from the reported benchmark.

The implemented closure and instability claim. On p. 4 the source specifies the finite local model

$$\partial_m q_{ijm} = \kappa v_{th}(p_{ij} - \delta_{ij}p), \qquad p = \mathrm{tr}(p_{ij})/3, \qquad \kappa = 0.1 d_e^{-1},$$

with an isotropization timescale of about $90\omega_{pe}^{-1}$ (original TeX line 179). Figure 3's caption, p. 4, explicitly omits its contribution from the displayed fluid heat flux and notes that only its divergence is constrained (line 166). The finite term is therefore a disclosed part of the model. The longitudinal calculation in this study tests that finite model, rather than silently replacing it by its zero-coefficient limit. On p. 5 the source also asserts that the two-electron-fluid model can capture linear growth because the instabilities are due to counter-streaming (line 207). The general importance of closure choice for instability fidelity is established prior context [7].

Accuracy metric and retained qualifications. The color in source Fig. 4, p. 5, is the best-time score

$$\max_t \left[1 - \frac{\langle |q_{xxx} - q_{CSxxx}| \rangle_{xy}}{\langle |q_{xxx}| \rangle_{xy}}\right]$$

(original TeX line 198). Its proposed thermal/drift scaling is explained in line 202. The present audit distinguishes this empirical best-time diagnostic from a uniform error guarantee; it does not

claim to have rerun the parameter scan. The source acknowledges later stream folding (p. 5, line 207), separate fluid color scales and amplitude differences in the full-tensor comparison (Fig. 5, p. 7, line 250), and kinetic free-expansion limitations (Appendix C, p. 8, line 293). The exact mixture identities in source Eqs. (3)–(4) and (B1)–(B2) are preserved as valid algebra.

B Pinned execution record

The reported calculations use the immutable code commit:

e1ebd1aaee17bc5cd486d40387eb655aefd70c20

This pins the numerical code and inputs; the final release also contains the manuscript source.

```
git clone https://github.com/Teylersf/counterstreaming-heatflux-audit.git
cd counterstreaming-heatflux-audit
git checkout e1ebd1aaee17bc5cd486d40387eb655aefd70c20
python -m pip install -r requirements.txt
python experiments/fetch_data.py
python experiments/data_checks.py
python -X utf8 experiments/verify.py
python -X utf8 experiments/make_figures.py
```

This is an author-environment execution record, not third-party certification or independent human peer review.

```
Started UTC: 2026-10-28T23:09:06.987425+00:00
Python: 3.12.14 (main, Aug 25 2026, 14:01:42) [MSC v.1944 64 bit (AMD64)]
NumPy: 2.3.5
h5py: 3.16.0
PASS: initial_current
PASS: initial_pressure
PASS: mixture_identities
PASS: continuity_symmetry
PASS: conditioning_counterexample
PASS: singlefluid_nonidentifiability
PASS: finite_relaxation_stability
PASS: isotropization_entropy
All assertions passed: true
Exit status: 0
```

File	SHA-256
results/verification.json	6e724e564d4ca23bf718d4add4eded9311aa6ebbeb161c4abc606ec750f01491
experiments/verify.py	23c64243de9e58d493fc87b18b4c47edf917fb67d11853898009e6e8824903e6
experiments/closure_checks.py	797fa1a2c85b36e09531fcff343ebed773674ecfe5a1caf8bd6518826ed3efbb
experiments/data_checks.py	fb2253e5f1970e80baea7a5349a8ac0cbf0e847b0cfe3f04eadb99b1e32e064b
experiments/fetch_data.py	9739bc225cda81134c984edb9c06a39a36a5c24403d6b67564e6de2d063a6fc0

The full log, input provenance, derived profiles and strict verification JSON are included in the repository.